

ITAI 1371 – Intro To Machine Learning

Prof. Viswanatha Rao

Marisabel Morales

W213048975

Lab 03 Reflective Journal – ML Workflow & Learning Types

This lab helped me understand the machine learning workflow much better because I was able to see how all the concepts connect instead of looking at them as separate steps. At the beginning, I understood the general idea of machine learning, but I was still confused about what training, testing, prediction, and evaluation actually meant. After working through the lab, the process became much more logical to me.

One of the most important things I learned was why the data needs to be separated into training and testing sets. I now understand that the training data is used to teach the model, while the testing data is information the model has not seen before. This is important because testing the model with the same information used for training would not really show whether it learned. Thinking about the test data as an exam helped me understand this concept. The model needs to be evaluated on something new to see if it can recognize the patterns it learned.

I also have a clearer understanding of supervised, unsupervised, and reinforcement learning. Supervised learning became easier for me when I thought about it as having an answer key during training. The model has examples and the correct labels. In unsupervised learning, there is no answer key, so the model has to look for patterns or groups by itself. Reinforcement learning is different because the model learns through actions and feedback, such as rewards or penalties.

Another part that was very useful was comparing Logistic Regression and Decision Tree using the Wine dataset. Logistic Regression originally achieved 88.9% accuracy, while the Decision Tree achieved 83.3%. Before this lab, I did not really understand why we would use different models with the same data. Now I understand that different algorithms can interpret patterns differently, so evaluating and comparing them is necessary.

The feature experiment was also one of the parts that impacted me the most. The original Logistic Regression model used four features and achieved 88.9% accuracy. I changed the features to alcohol, hue, and flavanoids, and the accuracy increased to 94.4%. This helped me see that the model itself is not the only thing affecting the result. The information we choose to give the model can also make a significant difference. It made me more curious about how we determine which features are actually the most useful.

I also completed an additional practice using the Breast Cancer dataset from Scikit-learn. I wanted to see if I could apply the same workflow to completely different data. Before training the model, I explored the dataset instead of immediately running the model. I checked its shape and learned that it contains 569 samples and 30 features. I also looked at the feature names, target classes, and class distribution. I converted the data into a Pandas DataFrame, which helped me see more clearly what information was actually inside the dataset. I found 357 benign and 212 malignant samples.

After exploring the data, I divided it into training and testing sets, trained a Logistic Regression model, made predictions, and evaluated the results. The model achieved 95.6% accuracy. What was most useful about this additional example was realizing that even though the dataset and subject were completely different, the basic workflow stayed almost the same.

I also learned more about the code itself during this exercise. For example, I now understand that `fit()` is used to train the model, `predict()` uses the trained model with the test data, and `accuracy_score()` compares those predictions with the actual answers. I also learned how Pandas can help me explore data before building a model.

The biggest change for me after this lab is that machine learning feels less like a group of complicated commands and more like a process that I can follow and understand. I still want to learn more about how to select the best features, how to choose between different models, and how to know whether a high accuracy score will continue to be reliable when the model receives completely new data.